1. **Запустила сервер и получила PID backend-процесса**

```
postgres=# SELECT pg_backend_pid
postgres-# ();
 pg_backend_pid
----------------
          30985
(1 row)
```

2. **Присоединила отладочную сборку к этому процессу в отдельном терминале**

```
user@MacBook-Air-user ~ % lldb -p 30985
(lldb) process attach --pid 30985
Process 30985 stopped
* thread #1, queue = 'com.apple.main-thread', stop reason = signal SIGSTOP
    frame #0: 0x00007ff819331822 libsystem_kernel.dylib`kevent + 10
libsystem_kernel.dylib`kevent:
->  0x7ff819331822 <+10>: jae    0x7ff81933182c ; <+20>
    0x7ff819331824 <+12>: movq   %rax, %rdi
    0x7ff819331827 <+15>: jmp    0x7ff81932d29a ; cerror_nocancel
    0x7ff81933182c <+20>: retq
Target 0: (postgres) stopped.
Executable binary set to "/Users/user/postgres/install/debug/bin/postgres".
Architecture set to: x86_64-apple-macosx-.
(lldb) continue
Process 30985 resuming
```

3. **Установила точку остановки внутри расширения и вызвала функцию**

```
(lldb) breakpoint set --name hello_world
Breakpoint 1: no locations (pending).
WARNING:  Unable to resolve breakpoint to any actual locations.
1 location added to breakpoint 1
Process 30985 stopped
* thread #1, queue = 'com.apple.main-thread', stop reason = breakpoint 1.1
    frame #0: 0x0000000107ebf46c demo_extension.dylib`hello_world(fcinfo=0x00007f
bf800247f0) at demo_extension.c:11:5
Target 0: (postgres) stopped.
```

4. **Использовала команду bt для просмотра стека вызовов**

```
(lldb) bt
* thread #1, queue = 'com.apple.main-thread', stop reason = breakpoint 1.1
  * frame #0: 0x0000000107ebf46c demo_extension.dylib`hello_world(fcinfo=0x00007fbf800247f0) at demo_extension.c:11:5
    frame #1: 0x000000010731445b postgres`ExecInterpExpr(state=0x00007fbf80024710, econtext=0x00007fbf80024e08, isnull=0x00007ff7b8f08447) at execExprInterp.c:934:8
    frame #2: 0x0000000107313167 postgres`ExecInterpExprStillValid(state=0x00007fbf80024710, econtext=0x00007fbf80024e08, isNull=0x00007ff7b8f08447) at execExprInterp.c:2307:9
    frame #3: 0x00000001074c3eeb postgres`ExecEvalExprSwitchContext(state=0x00007fbf80024710, econtext=0x00007fbf80024e08, isNull=0x00007ff7b8f08447) at executor.h:440:13
    frame #4: 0x00000001074c3dd1 postgres`evaluate_expr(expr=0x00007fbf8000f260, result_type=25, result_typmod=-1, result_collation=100) at clauses.c:5011:14
    frame #5: 0x00000001074c5af5 postgres`evaluate_function(funcid=16388, result_type=25, result_typmod=-1, result_collid=100, input_collid=0, args=0x0000000000000000, funcvariadic=false, func_tuple=0
x00007fbf7fab3840, context=0x00007ff7b8f09d78) at clauses.c:4518:9
    frame #6: 0x00000001074c50df postgres`simplify_function(funcid=16388, result_type=25, result_typmod=-1, result_collid=100, input_collid=0, args_p=0x00007ff7b8f08a20, funcvariadic=false, process_ar
gs=true, allow_non_const=true, context=0x00007ff7b8f09d78) at clauses.c:4107:12
    frame #7: 0x00000001074c0af2 postgres`eval_const_expressions_mutator(node=0x00007fbf8000eb08, context=0x00007ff7b8f09d78) at clauses.c:2592:14
    frame #8: 0x00000001073eb0a0 postgres`expression_tree_mutator_impl(node=0x00007fbf8000eb58, mutator=(postgres`eval_const_expressions_mutator at clauses.c:2443), context=0x00007ff7b8f09d78) at node
Funcs.c:3482:5
    frame #9: 0x00000001074c2f83 postgres`eval_const_expressions_mutator(node=0x00007fbf8000eb58, context=0x00007ff7b8f09d78) at clauses.c:3726:9
    frame #10: 0x00000001073eb4b2 postgres`expression_tree_mutator_impl(node=0x00007fbf8000eba8, mutator=(postgres`eval_const_expressions_mutator at clauses.c:2443), context=0x00007ff7b8f09d78) at nod
eFuncs.c:3568:12
    frame #11: 0x00000001074c2f83 postgres`eval_const_expressions_mutator(node=0x00007fbf8000eba8, context=0x00007ff7b8f09d78) at clauses.c:3726:9
    frame #12: 0x00000001074c04da postgres`eval_const_expressions(root=0x00007fbf8000ecc8, node=0x00007fbf8000eba8) at clauses.c:2267:9
    frame #13: 0x000000010749 4c98 postgres`preprocess_expression(root=0x00007fbf8000ecc8, expr=0x00007fbf8000eba8, kind=1) at planner.c:1354:10
    frame #14: 0x000000010749 38f8 postgres`subquery_planner(glob=0x00007fbf8000e238, parse=0x00007fbf8000e348, parent_root=0x0000000000000000, hasRecursion=false, tuple_fraction=0, setops=0x0000000000
000000) at planner.c:924:3
    frame #15: 0x000000010749285d postgres`standard_planner(parse=0x00007fbf8000e348, query_string="SELECT hello_world();", cursorOptions=2048, boundParams=0x0000000000000000) at planner.c:453:9
    frame #16: 0x000000010749 25c3 postgres`planner(parse=0x00007fbf8000e348, query_string="SELECT hello_world();", cursorOptions=2048, boundParams=0x0000000000000000) at planner.c:313:12
    frame #17: 0x000000010760b13c postgres`pg_plan_query(querytree=0x00007fbf8000e348, query_string="SELECT hello_world();", cursorOptions=2048, boundParams=0x0000000000000000) at postgres.c:901:9
    frame #18: 0x000000010760b32a postgres`pg_plan_queries(querytrees=0x00007fbf8000ec78, query_string="SELECT hello_world();", cursorOptions=2048, boundParams=0x0000000000000000) at postgres.c:995:11
    frame #19: 0x000000010760e48b postgres`exec_simple_query(query_string="SELECT hello_world();") at postgres.c:1193:19
    frame #20: 0x000000010760d7af postgres`PostgresMain(dbname="postgres", username="user") at postgres.c:4770:7
    frame #21: 0x0000000107605f88 postgres`BackendMain(startup_data=0x00007ff7b8f0a568, startup_data_len=24) at backend_startup.c:124:2
    frame #22: 0x00000001075040dd postgres`postmaster_child_launch(child_type=B_BACKEND, child_slot=1, startup_data=0x00007ff7b8f0a568, startup_data_len=24, client_sock=0x00007ff7b8f0a5f0) at launch_b
ackend.c:290:3
    frame #23: 0x000000010750a8a5 postgres`BackendStartup(client_sock=0x00007ff7b8f0a5f0) at postmaster.c:3569:8
    frame #24: 0x0000000107508e85 postgres`ServerLoop at postmaster.c:1703:6
    frame #25: 0x0000000107507c48 postgres`PostmasterMain(argc=3, argv=0x00007fbf7f904cd0) at postmaster.c:1401:11
    frame #26: 0x00000001073b5ab3 postgres`main(argc=3, argv=0x00007fbf7f904cd0) at main.c:227:4
    frame #27: 0x00007ff818fc9530 dyld`start + 3056
```

Командой expression fcinfo просмотрела структуру аргументов

```
(lldb) expression *fcinfo
(FunctionCallInfoBaseData) $0 = {
  flinfo = 0x00007fbf800247a0
  context = NULL
  resultinfo = NULL
  fncollation = 0
  isnull = false
  nargs = 0
  args = {}
}
```

Поставила watchpoint на fcinfo->resultinfo

```
(lldb) watchpoint set expression -- fcinfo->resultinfo
Watchpoint created: Watchpoint 1: addr = 0x00000000 size = 8 state = enabled type = m
    watchpoint spec = 'fcinfo->resultinfo'
    watchpoint resources:
       #0: addr = 0x00000000 size = 8
Watchpoint 1 hit:

new value: error: read memory from 0x0 failed (0 of 8 bytes read)
(lldb) continue
Process 30985 resuming
```

5. **Поставила точку остановки на функцию ядра cstring_to_text**

```
(lldb) breakpoint set --name cstring_to_text
Breakpoint 2: where = postgres`cstring_to_text + 12 at varlena.c:195:34, address = 0x00000001077dc1cc
Process 30985 stopped
* thread #1, queue = 'com.apple.main-thread', stop reason = breakpoint 1.1
    frame #0: 0x0000000107ebf46c demo_extension.dylib`hello_world(fcinfo=0x00007fbf800247f0) at demo_extension.c:11:5
Target 0: (postgres) stopped.
(lldb) continue
Process 30985 resuming
Process 30985 stopped
* thread #1, queue = 'com.apple.main-thread', stop reason = breakpoint 2.1
    frame #0: 0x00000001077dc1cc postgres`cstring_to_text(s="Hello, World from extension!") at varlena.c:195:34
   192  text *
   193  cstring_to_text(const char *s)
   194  {
-> 195          return cstring_to_text_with_len(s, strlen(s));
   196  }
   197
   198  /*
Target 0: (postgres) stopped.
```